

On run statistics for binary trials

Valeri T. Stefanov¹

Department of Mathematics and Statistics, The University of Western Australia, Nedlands 6907, W.A., Australia

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Abstract

The distribution theory of run statistics, both exact and asymptotic, is important in applied probability and statistical inference. In this paper we derive the asymptotic normal distribution of a whole collection of run statistics, which count failure and success runs taking into account their length, in Markov-dependent Bernoulli trials. From the viewpoint of applications, our expressions for the limiting parameters require minimal computational effort. © 2000 Elsevier Science B.V. All rights reserved.

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1. Introduction

The distribution theory of run statistics, exact and asymptotic, for both independent and dependent Bernoulli trials, is important in applied probability and statistical inference. In the present paper we are concerned with the asymptotic theory. Very early contributions are due to Mood (1940), Levene and Wolfowitz (1944), Wolfowitz (1944), Koopman (1950) and Dobrushin (1953). Most of the recent literature concerns one-dimensional run statistics (see Wang and Ji (1995) and the references therein) and the associated limit results are primarily Poisson-type approximations. An exception is the result of Jennen-Steinmetz and Gasser (1986) where the limiting distribution is multivariate normal. They treat independent nonidentically (but “close” to identically) distributed Bernoulli trials and cover the joint distribution of a whole collection of run statistics which count the success and failure runs taking into account their length. Such results are of importance in hypothesis testing theory, in particular for developing tests for randomness, because runs together with their length utilize much more sample information than most of the available test statistics based on runs. The result of

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Jennen-Steinmetz and Gasser (1986) is an extension of an earlier result due to Mood (1940) and is in the spirit of Levene and Wolfowitz (1944) and Wolfowitz (1944). In the present paper we extend the asymptotic results of Jennen-Steinmetz and Gasser (1986) to some dependent Bernoulli trials; in particular, our result covers the cases of independent identically distributed trials and first- and second-order Markov-dependent trials. The methodology is based on selecting a suitable finite-state Markov chain, imbedding the problem into it, viewing the chain as a member of a special exponential family of stochastic processes, and elaborating on an appropriate explicit limit result from Stefanov (1995). The Markov chain imbedding technique is natural for studying runs related problems and has been used by many authors during the last two decades (cf. the references in Fu and Koutras, 1994; Stefanov and Pakes, 1997). The exponential family technology has been recently introduced to these problems in Stefanov and Pakes (1997). Note that no exact result is available for the joint distribution of the aforementioned collection of run statistics. Exact results for one- or two-dimensional distributions, concerning the total number of success and/or failure runs and the longest run, are provided in Fu and Koutras (1994), Fu (1996) and Lou (1996).

2. The main result

There are many ways of counting runs in sequences of binary trials. The two outcomes in binary trials are conventionally called “success” (“1”) and “failure” (“0”). Throughout the paper, runs are understood in the sense of Mood’s (1940) counting rules, unless otherwise explicitly stated. For example, the sequence

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has the following failure runs: one of length one, two of length two and one of length three, and success runs: two of length two and one of length three. For other ways of counting runs see, for example, Wang and Ji (1995). It is a well-known fact that the progress of achieving success runs, in independent, or Markov-dependent, Bernoulli trials, can be imbedded in the so-called “success run” Markov chain (see Karlin, 1975, p. 54). An entry to a state, say $r, r \geq 1$, marks achieving a success run of length at least r . Such a Markov chain accounts for success runs and their length and the total number of failure runs, but it does not account for the length of failure runs. A modification of the “success run” Markov chain can accommodate the latter. Therefore, we introduce the following special finite-state Markov chain and three motivating examples.

Let $(Z(t); t=0, 1, \dots)$ be a Markov chain with $k+m$ states, denoted by $1, 2, \dots, k+m$, whose one-step transition probability matrix is given by $(p_{i,j}, i, j = 1, 2, \dots, k+m)$. Furthermore the only nonzero entries of this matrix are

$$\begin{aligned} p_{i,i+1}, & \quad i = 1, 2, \dots, k+m-1, \\ p_{i,k+1}, & \quad i = 1, 2, \dots, k, \end{aligned}$$

$$p_{i,1}, \quad i = k+1, k+2, \dots, k+m,$$

$$p_{k,k}$$

and

$$p_{k+m,k+m}.$$

We shall assume $Z(0) = 1$. Denote by $N_{i,j}(t)$ the number of one-step transitions from state i to state j up to time t . Let

$$\tau_n = \inf\{t: N_{k+m,1}(t) = n\}, \quad n = 1, 2, \dots$$

For brevity and clarity we shall use the following notation for $N_{i,j}$'s and $p_{i,j}$'s:

$$R_i(\tau_n) = N_{i,i+1}(\tau_n) \quad \text{if } i = 1, 2, \dots, k-1, k+1, k+2, \dots, k+m-1$$

$$R_k(\tau_n) = N_{k,k}(\tau_n),$$

$$R_{k+m}(\tau_n) = N_{k+m,k+m}(\tau_n),$$

$$L_i(\tau_n) = N_{i,k+1}(\tau_n) \quad \text{if } i = 1, 2, \dots, k,$$

$$L_i(\tau_n) = N_{i,1}(\tau_n) \quad \text{if } i = k+1, k+2, \dots, k+m,$$

$$p_i = p_{i,i+1}, \quad i = 1, 2, \dots, k-1, k+1, k+2, \dots, k+m-1,$$

$$p_k = p_{k,k},$$

$$p_{k+m} = p_{k+m,k+m},$$

$$q_i = p_{i,k+1}, \quad i = 1, 2, \dots, k,$$

$$q_i = p_{i,1}, \quad i = k+1, k+2, \dots, k+m,$$

and even more briefly R_i and L_i will stand for $R_i(\tau_1)$ and $L_i(\tau_1)$ respectively. Clearly $p_i + q_i = 1$ for each $i = 1, 2, \dots, k+m$.

Example 1 (Independent Bernoulli trials). Denote by τ the time of first seeing at least m consecutive successes followed by a failure, that is the time until first seeing either of the following patterns:

$$\underbrace{11 \dots 1}_j 0 \quad j = m, m+1, \dots$$

in independent Bernoulli trials with a probability of success p . Then it is easy to see that $\tau_1 = \tau$ if we substitute p_i 's and q_i 's as follows:

$$p = q_1 = q_2 = \dots = q_k = p_{k+1} = p_{k+2} = \dots = p_{k+m},$$

$$q = p_1 = p_2 = \dots = p_k = q_{k+1} = q_{k+2} = \dots = q_{k+m}.$$

Note also that the R_i 's have the following interpretation: for $i = 2, 3, \dots, k$, R_i is the number of failure runs of length at least i , and for $i = k+1, k+2, \dots, k+m$, R_i is the number of success runs of length at least $i - k + 1$, all counted up to time τ_1 . Also $L_1 + L_2 + \dots + L_k$ (later in the paper we show that this quantity equals $L_1 + R_1$) is the number of all success runs.

Example 2 (*Markov-dependent Bernoulli trials*). We obtain the Markov-dependent version of the preceding example if

$$\begin{aligned} p_{00} &= p_1 = p_2 = \cdots = p_k, \\ p_{01} &= q_1 = q_2 = \cdots = q_k, \\ p_{11} &= p_{k+1} = p_{k+2} = \cdots = p_{k+m}, \\ p_{10} &= q_{k+1} = q_{k+2} = \cdots = q_{k+m}, \end{aligned}$$

where p_{ij} , $i, j = 0, 1$, are the transition probabilities of a two-state Markov chain with state 0 identified as a “failure” and state 1 as a “success”. The initial state in this case is assumed to be 0.

Example 3 (*Second-order Markov-dependent Bernoulli trials*). The second-order Markov-dependent version of the same example is derived if

$$\begin{aligned} p_{100} &= p_1, \\ p_{101} &= q_1, \\ p_{000} &= p_2 = p_3 = \cdots = p_k, \\ p_{001} &= q_2 = q_3 = \cdots = q_k, \\ p_{011} &= p_{k+1}, \\ p_{010} &= q_{k+1}, \\ p_{111} &= p_{k+2} = p_{k+3} = \cdots = p_{k+m}, \\ p_{110} &= q_{k+2} = q_{k+3} = \cdots = q_{k+m}, \end{aligned}$$

where p_{ijr} , $i, j, r = 0, 1$, is the probability of visiting state r given that a step back the process visited state j and two steps back it visited state i . We assume that the initial two steps of the process are first 1 (“success”) then 0 (“failure”).

The interpretation of the R_i ’s in the preceding two examples is the same as that in Example 1.

Remark 1. The choice of the initial state in the above examples does not affect the asymptotic results we derive in this paper; they hold for any initial distribution of the associated Markov chains.

The Radon–Nikodym derivative of the measure generated by the chain on the time interval $[0, \tau_n]$, with respect to some σ -finite measure is (Stefanov, 1991)

$$\exp \left\{ \sum_{i,j=1}^{k+m} [N_{i,j}(\tau_n) \ln p_{i,j}] \right\} = \exp \left\{ \sum_{i=1}^{k+m} [R_i(\tau_n) \ln p_i + L_i(\tau_n) \ln q_i] \right\} \quad (2.1)$$

If we view n as a time-parameter then (2.1) represents an exponential family of stochastic processes (cf. Küchler and Sørensen, 1997). In view of Stefanov (1991, Proposition 1) the family given by (2.1) is a noncurved exponential family of order $(k+m)$. Refer to Barndorff-Nielsen (1978, 1980) and Brown (1986) for formal definitions of the

relevant terms associated with exponential families, and Küchler and Sørensen (1997) for those concerning exponential families of stochastic processes.

We shall find a minimal canonical representation of the family given by (2.1). In view of Stefanov (1991, see (3), p. 355) we get the following linear relationships:

$$\begin{aligned} R_1 + L_1 &= L_{k+1} + L_{k+2} + \cdots + L_{k+m}, \\ R_{i+1} + L_{i+1} &= R_i \quad \text{if } i = 1, 2, \dots, k-1, \\ L_k &= R_{k-1}, \\ L_{k+1} + R_{k+1} &= L_1 + L_2 + \cdots + L_k, \\ R_{i+1} + L_{i+1} &= R_i \quad \text{if } i = k+1, k+2, \dots, k+m-2, \\ L_{k+m} &= 1, \\ R_{k+m-1} &= 1. \end{aligned}$$

Actually these are derived by counting the number of entries to, and exits from, a state; note that for each state these two quantities are equal because the initial and final state is 1. Using these relations one can easily see that all L_i 's can be expressed linearly in terms of $L_1, R_1, R_2, \dots, R_{k+m-2}, R_{k+m}$. Therefore one arrives at the following minimal canonical representation of the exponential family given by (2.1) when $n=1$:

$$\exp \left\{ \sum_{i=1}^{k+m} [\theta_i X_i] - \varphi(\theta) \right\},$$

where $X_1 = L_1$, $X_i = R_{i-1}$, $i = 2, 3, \dots, k+m-1$, $X_{k+m} = R_{k+m}$ and

$$\begin{aligned} \theta_1 &= \ln(q_1 q_{k+1}), \\ \theta_2 &= \ln(p_1 q_2 q_{k+1}), \\ \theta_i &= \ln(p_{i-1} q_i / q_{i-1}) \quad \text{if } i = 3, 4, \dots, k, k+2, \dots, k+m-1, \\ \theta_{k+1} &= \ln(p_k), \\ \theta_{k+m} &= \ln(p_{k+m}), \end{aligned} \tag{2.2}$$

with the implicit expression for $\varphi(\theta) = -\ln(q_{k+m} p_{k+m-1} / q_{k+m-1})$. Similarly the minimal canonical representation of (2.1) for any n is

$$\exp \left\{ \sum_{i=1}^{k+m} [\theta_i X_i(\tau_n)] - n\varphi(\theta) \right\},$$

where $X_1(\tau_n) = L_1(\tau_n)$, $X_i(\tau_n) = R_{i-1}(\tau_n)$, $i = 2, 3, \dots, k+m-1$, $X_{k+m}(\tau_n) = R_{k+m}(\tau_n)$ and note that $L_{k+m}(\tau_n) = R_{k+m-1}(\tau_n) = n$.

From (2.2) we can find an explicit expression for $\varphi(\cdot)$ in terms of the θ_i 's as follows. Observe first that

$$\frac{q_2}{p_2} = \frac{e^{\theta_2}}{q_{k+1} - e^{\theta_1} - e^{\theta_2}}.$$

Since

$$q_i = \frac{e^{\theta_i} q_{i-1}}{p_{i-1}} \quad (2.3)$$

for each i , $i = 3, 4, \dots, k-1$, and $i = k+2, k+3, \dots, k+m-1$, we derive

$$q_k = \frac{e^{\theta_2 + \theta_3 + \dots + \theta_k}}{q_{k+1} - e^{\theta_1} - e^{\theta_2} - e^{\theta_2 + \theta_3} - e^{\theta_2 + \theta_3 + \theta_4} - \dots - e^{\theta_2 + \theta_3 + \dots + \theta_{k-1}}}$$

which together with

$$q_k = 1 - e^{\theta_{k+1}}$$

yield

$$q_{k+1} = \frac{(1 - e^{\theta_{k+1}})(e^{\theta_1} + \sum_{i=2}^{k-1} e^{\theta_2 + \theta_3 + \dots + \theta_i}) + e^{\theta_2 + \theta_3 + \dots + \theta_k}}{1 - e^{\theta_{k+1}}}$$

and

$$\frac{q_{k+1}}{p_{k+1}} = \frac{(1 - e^{\theta_{k+1}})(e^{\theta_1} + \sum_{i=2}^{k-1} e^{\theta_2 + \theta_3 + \dots + \theta_i}) + e^{\theta_2 + \theta_3 + \dots + \theta_k}}{(1 - e^{\theta_{k+1}})(1 - e^{\theta_1} - \sum_{i=2}^{k-1} e^{\theta_2 + \theta_3 + \dots + \theta_i}) - e^{\theta_2 + \theta_3 + \dots + \theta_k}}. \quad (2.4)$$

For brevity the right hand side in (2.4) will be denoted by $C(\theta_1, \theta_2, \dots, \theta_k)$ and even more briefly by C . Further in view of (2.2) we find that

$$\frac{q_{k+m-1}}{p_{k+m-1}} = \frac{C e^{\theta_{k+2} + \theta_{k+3} + \dots + \theta_{k+m-1}}}{1 - C \sum_{i=k+2}^{k+m-1} e^{\theta_{k+2} + \theta_{k+3} + \dots + \theta_i}}$$

which together with

$$q_{k+m} = 1 - e^{\theta_{k+m}}$$

imply

$$\varphi(\theta) = \ln \frac{C e^{\theta_{k+2} + \theta_{k+3} + \dots + \theta_{k+m-1}}}{(1 - e^{\theta_{k+m}})[1 - C \sum_{i=k+2}^{k+m-1} e^{\theta_{k+2} + \theta_{k+3} + \dots + \theta_i}]}, \quad (2.5)$$

where C has been introduced above.

From well-known properties of exponential families (cf. Barndorff-Nielsen, 1978; p. 114 or Küchler and Sørensen, 1997, p. 104) we have the following expressions for the expectation vector, EX , and covariance matrix, Σ , of the random vector $X = (X_1, X_2, \dots, X_{m+k})$: (recall $X_i = X_i(\tau_1)$)

$$EX = \nabla \varphi(\theta) = \left(\frac{\partial \varphi}{\partial \theta_1}, \frac{\partial \varphi}{\partial \theta_2}, \dots, \frac{\partial \varphi}{\partial \theta_{k+m}} \right)^T \quad \text{and} \quad \Sigma = \left(\frac{\partial^2 \varphi}{\partial \theta_i \partial \theta_j} \right)_{i,j=1}^{k+m}.$$

Note that

$$\tau_1 = 2X_1 + 3X_2 + \sum_{i=3}^{k+m} X_i + 1$$

because

$$\tau_1 = \sum_{i,j} N_{i,j}(\tau_1)$$

and

$$\sum_{i=2}^k L_i = X_2, \quad \sum_{i=k+1}^{k+m} L_i = X_1 + X_2, \quad R_{k+m-1} = 1.$$

The Markov chain regenerates at the τ_i 's and therefore the vector process $X(\tau_n)$, $n = 1, 2, \dots$, has stationary independent increments. Also the expectation vector and covariance matrix of the increment $X(\tau_{n+1}) - X(\tau_n)$ are equal to EX and Σ , respectively. The latter statements follow, for example, from Theorem 1 of Stefanov (1986). Note that

$$\tau_n = 2X_1(\tau_n) + 3X_2(\tau_n) + \sum_{i=3}^{k+m} X_i(\tau_n) + 1.$$

From the central limit theorem for i.i.d. random vectors we have

$$\frac{X(\tau_n) - n\nabla\varphi(\theta)}{\sqrt{n}} \Rightarrow N(0, \Sigma) \quad (2.6)$$

as $n \uparrow +\infty$.

This is a limit result for the quantities of interest when counted up to the times τ_i . Now some general limit results in Stefanov (1995) are applicable here in order to pass from the τ_i 's to nonrandom epochs of time. This works as follows. Introduce first the following notation:

$$\delta(\theta) = 2\frac{\partial\varphi(\theta)}{\partial\theta_1} + 3\frac{\partial\varphi(\theta)}{\partial\theta_2} + \frac{\partial\varphi(\theta)}{\partial\theta_3} + \dots + \frac{\partial\varphi(\theta)}{\partial\theta_{k+m}} + 1,$$

$$M_1 = \begin{bmatrix} 1 & 1 & 0 & \dots & \dots & 0 \\ 0 & 1 & 0 & \dots & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \dots & \dots & 0 & 1 & 0 \\ 2 & 3 & 1 & \dots & 1 & 1 \end{bmatrix},$$

$$(l_1, l_2, \dots, l_{k+m})^T = (\delta(\theta))^{-1} M_1 \nabla\varphi(\theta),$$

$$M_2 = \begin{bmatrix} 1 & 0 & \dots & \dots & 0 & l_1 \\ 0 & 1 & 0 & \dots & 0 & l_2 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \dots & \dots & 0 & 1 & l_{k+m-1} \end{bmatrix},$$

where $(\cdot)^T$ denotes the transpose of $(\cdot)$, and M_1 and M_2 are $(k+m) \times (k+m)$ and $(k+m-1) \times (k+m)$ matrices respectively; note that $l_{k+m} = 1$.

Then (2.6) implies

$$\frac{Y(\tau_n) - nM_1 \nabla \varphi(\theta)}{\sqrt{n}} \Rightarrow N(0, M_1 \Sigma M_1^T), \quad (2.7)$$

where $Y(\tau_n) = M_1 X(\tau_n)$; note that $Y_i(\tau_n) = X_i(\tau_n)$ if $i = 2, 3, \dots, k + m - 1$ and $Y_1(\tau_n) = X_1(\tau_n) + X_2(\tau_n)$ and $Y_{k+m}(\tau_n) = \tau_n - 1$. Also note that

$$E\tau_1 = \delta(\theta)$$

and

$$M_2 Y(\tau_n) = (Y_1(\tau_n), Y_2(\tau_n), \dots, Y_{k+m-1}(\tau_n))^T - (\tau_n - 1)(l_1, l_2, \dots, l_{k+m-1})^T.$$

Then in view of (2.7) we have

$$\frac{(Y_1(\tau_n), \dots, Y_{k+m-1}(\tau_n))^T - (\tau_n - 1)(l_1, \dots, l_{k+m-1})^T}{\sqrt{n}} \Rightarrow N(0, M_2 M_1 \Sigma M_1^T M_2^T),$$

and due to an asymptotic equivalence

$$\frac{(Y_1(\tau_n), \dots, Y_{k+m-1}(\tau_n))^T - \tau_n(l_1, \dots, l_{k+m-1})^T}{\sqrt{n}} \Rightarrow N(0, M_2 M_1 \Sigma M_1^T M_2^T), \quad (2.8)$$

where note that $E(Y_i(\tau_n) - \tau_n l_i) = 0$ for each $i, i = 1, 2, \dots, k + m - 1$.

Recall the interpretation of the R_i 's from Example 1 and the implied one for the $Y_i(\tau_n)$'s, and similarly let $Y_i(t)$ be the counterparts of the latter for a nonrandom t , that is, $Y_i(t)$, $i = 2, 3, \dots, k$, is the number of runs of failures of length at least i , $Y_i(t)$, $i = k + 1, k + 2, \dots, k + m$, is the number of runs of successes of length at least $i - k + 1$, and $Y_1(t)$ is the number of all runs of successes, in t binary trials of either of the three types as introduced in Examples 1, 2 and 3. Thus, the first $k + m - 1$ components of $Y(t)$ contain the number of success runs of length at least i for each $i = 1, 2, \dots, m$, and the number of failure runs of length at least j for each $j = 2, 3, \dots, k$. Note that the number of all success runs is asymptotically equivalent to the number of all failure runs; therefore we need only one of these two quantities. Now (2.8) together with a general result in Stefanov (1995, Proposition 7.3) yield the following.

Theorem 1. *The joint distribution of $Y_1(t), Y_2(t), \dots, Y_{k+m-1}(t)$ is asymptotically normal with*

$$\frac{(Y_1(t), \dots, Y_{k+m-1}(t))^T - t(l_1, \dots, l_{k+m-1})^T}{\sqrt{t}} \Rightarrow N(0, (\delta(\theta))^{-1} M_2 M_1 \Sigma M_1^T M_2^T),$$

as $t \uparrow +\infty$, where all quantities expressed in terms of the θ_i 's are to be evaluated in terms of the transition probabilities using (2.2).

Remark 2. The above theorem concerns runs in Mood's counting rules. Note that simple linear transformations of the Y_i 's lead to the same run statistics when using either of the available counting rules of runs (cf. Wang and Ji 1995 for other counting rules). Therefore, from Theorem 1 we can derive the asymptotic normal distribution of the same quantities if either of those rules is used. Also in view of Proposition 7.3

of Stefanov (1995) the reader can easily state functional limit versions of the above theorem.

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